The argument, step by step — Step 4: realistic loudness

Fourth notebook; **one toggle — `loudness_structure`**. No new noise anywhere. What changes is honesty about *which combs a real machine gets to use*: step 1 granted the fixed pulsar its 30 strongest ruler combs regardless of quality; the campaign machine admits a comb only at $_{\mathrm{P}}$ and then keeps the best $_{\max}$ by information — because in a real analysis a $_{\mathrm{P}}$ comb cannot be attributed, templated, or trusted.

Under the V1 population's heavy-tailed loudness, that discipline collapses the *effective* number of combs, $_{\mathrm{eff}}$ $^{2\ 2}$ ^{4}, to the few winners — the campaign's banked $_{\mathrm{eff}}$ is the full machine's version of this number.

In [1]:
```
import numpy as np, matplotlib.pyplot as plt
import warnings; warnings.filterwarnings("ignore")
import hasasia, hasasia.sim as hsim, hasasia.sensitivity as hsen
np.seterr(all="ignore")
plt.rcParams.update({"figure.dpi":110,"font.size":9,"axes.spines.top":False,
                     "axes.spines.right":False,"legend.frameon":False})
BLUE, RED, INK = "#2b5aa0", "#a03030", "0.35"
print("hasasia", hasasia.__version__)

PARAMS=dict(SEED=20260821, L_TRUE=1.42, SIGMA_D=2.0,
            N_PSR=120, SIGMA_TOA=100e-9, SIGMA_PSR=450e-9, CAD_D=14.0,
            SPANS=(5,10,15,20,25), RHO_DET=5.0, NSRC=3000,
            F_POP=(1e-9,3.0e-7), NREAL=200,
            C_PC_S=2.99792458e8/3.0856775814913673e16, YR_S=365.25*86400.0)
P=PARAMS; TOOTH=0.25
WORLD=dict(red_noise=False, gwb=False, confusion=False, loudness_structure=T
print("WORLD toggles:", WORLD)

rng0=np.random.default_rng(P["SEED"])
freqs=np.logspace(np.log10(6e-10),-7,120)
_r=np.random.default_rng(42)
_phi=_r.uniform(0,2*np.pi,P["N_PSR"]); _th=np.arccos(_r.uniform(-1,1,P["N_PS
_CURVES={}
def curve(T):
    if T not in _CURVES:
        ps=hsim.sim_pta(timespan=float(T),cad=365.25/P["CAD_D"],sigma=P["SIG
                        phi=_phi,theta=_th,A_gwb=0,freqs=freqs)
        _CURVES[T]=hsen.DeterSensitivityCurve(
            [hsen.Spectrum(p,freqs=freqs) for p in ps]).h_c
    return _CURVES[T]

# ---- Faithful standalone port of active_wheel_demo_v4 cell 3 (V1): the NG1
# full-physics joint (Mc,z) population, ABSOLUTE normalisation, discrete tai
G_SI,C_SI=6.674e-11,2.998e8
MSUN,MPC=1.989e30,3.086e22
```

```
YRV=3.15576e7
T_YR=30; T_OBS=T_YR*YRV; F_YR=1/YRV
ALPHA_F=-8/3; F_RANGE=(2e-10,1e-7); F_BEND=2e-9; MC_MIN=1e8*MSUN
A_NG15=6.4e-15; RATE_RESIDUAL=2.47
NG_PSI0,NG_MPSI0,NG_MU,NG_EPS=-2.0,11.4,8.60,0.30
NG_APSI0,NG_AMU,NG_FBULGE=-1.21,1.10,0.615
NG_P0,NG_BETP=0.033,1.0
NG_T0GYR,NG_BETT,NG_GAMT=0.5,-0.5,-1.0
NG_PSIZ,NG_MPSIZ,NG_APSIZ=-0.60,0.11,-0.03
KH=(8/np.sqrt(10))*G_SI**(5/3)*np.pi**(2/3)/C_SI**4
freqsV=np.logspace(np.log10(F_RANGE[0]),-7,100)
H0_SI=69.33*1e3/MPC
_zg=np.linspace(0,3.0,3001)
_Ez=np.sqrt(0.228*(1+_zg)**3+0.772)
_dcg=np.concatenate([[0],np.cumsum((C_SI/H0_SI)*0.5*(1/_Ez[1:]+1/_Ez[:-1])*r
def d_com_m(z): return np.interp(z,_zg,_dcg)
def dt_dz(z): return 1.0/((1+z)*H0_SI*np.interp(z,_zg,_Ez))
_rngJ=np.random.default_rng(1234)
_lm1=np.linspace(9.5,13.2,190); _qs=np.linspace(0.25,1.0,26); _zs=np.linspac
_LM1,_QS,_ZS=np.meshgrid(_lm1,_qs,_zs,indexing="ij")
_dlm,_dq,_dz=_lm1[1]-_lm1[0],_qs[1]-_qs[0],_zs[1]-_zs[0]
_Psi=np.log(10)*10**(NG_PSI0+NG_PSIZ*_ZS) \
     *(10**_LM1/10**(NG_MPSI0+NG_MPSIZ*_ZS))**(1+NG_APSI0+NG_APSIZ*_ZS) \
     *np.exp(-10**_LM1/10**(NG_MPSI0+NG_MPSIZ*_ZS))
_Pp=NG_P0*(1+_ZS)**NG_BETP
_Tg=NG_T0GYR*1e9*YRV*(1+_ZS)**NG_BETT*_QS**NG_GAMT
_eta=_Psi*_dlm*(_Pp/_Tg)*dt_dz(_ZS)*_dz*_dq
_kp=_eta>0
_LM1,_QS,_ZS,_eta=_LM1[_kp],_QS[_kp],_ZS[_kp],_eta[_kp]
_NSC=16
_LM1=np.repeat(_LM1,_NSC); _QS=np.repeat(_QS,_NSC); _ZS=np.repeat(_ZS,_NSC)
_eta=np.repeat(_eta/_NSC,_NSC)
_M1=10**(NG_MU+NG_AMU*np.log10(NG_FBULGE*10**_LM1/1e11)+_rngJ.normal(0,NG_EP
_qbh=_QS**NG_AMU
SRC_MC=(_M1*(_M1*_qbh))**0.6/(_M1*(1+_qbh))**0.2*MSUN
_kp=SRC_MC>=MC_MIN
SRC_MC=SRC_MC[_kp]; SRC_Z=_ZS[_kp]; _eta=_eta[_kp]
_dc_m=d_com_m(SRC_Z); SRC_DL=(1+SRC_Z)*_dc_m
_Ctau=(5/96)*(G_SI*SRC_MC/C_SI**3)**(-5/3)*np.pi**(-8/3)
_shell=4*np.pi*(C_SI/MPC)*(1+SRC_Z)*(_dc_m/MPC)**2
SRC_W=RATE_RESIDUAL*_eta*_Ctau*(1+SRC_Z)**(-8/3)*_shell
SRC_AMP=KH*(SRC_MC*(1+SRC_Z))**(5/3)/SRC_DL
bin_edges=np.arange(F_RANGE[0],F_RANGE[1],1/T_OBS)
bin_cent=0.5*(bin_edges[1:]+bin_edges[:-1])
n_per_bin=SRC_W.sum()*np.maximum(bin_cent,F_BEND)**ALPHA_F/(bin_cent*T_OBS)
_ordA=np.argsort(SRC_AMP); _ampS=SRC_AMP[_ordA]
_cwS=np.cumsum(SRC_W[_ordA]); _cwT=_cwS[-1]
# split & UL curves (V1 recipe: 200-psr deepest bar/10; H95 anchored shape)
def _raw(n,s,t,A,seed=42,cad=26):
    r=np.random.default_rng(seed)
    ps=hsim.sim_pta(timespan=t,cad=cad,sigma=s*1e-9,phi=r.uniform(0,2*np.pi,
                    theta=np.arccos(r.uniform(-1,1,n)),A_gwb=A,freqs=freqsV)
    return hsen.DeterSensitivityCurve([hsen.Spectrum(p,freqs=freqsV) for p i
SNR_BAR=3.0
def det_curve(hc,bar=SNR_BAR): return bar*hc/np.sqrt(freqsV*T_OBS)
H_SPLIT=det_curve(_raw(200,100.0,T_YR,A_NG15))*10**(-0.5)/10.0
```

```
def h_split_of_f(f): return np.interp(f,freqsV,H_SPLIT)
_shape=_raw(67,100.0,15.0,A_NG15)/np.sqrt(freqsV*15*YRV)
H95=_shape*(8.0e-15/np.interp(6e-9,freqsV,_shape))
def h95_of_f(f): return np.interp(f,freqsV,H95)
def draw_population(seed=3):
    rng=np.random.default_rng(seed)
    hs=h_split_of_f(bin_cent)
    F,H,MCx=[],[],[]
    for k,fc in enumerate(bin_cent):
        amp_cut=hs[k]/fc**(2/3)
        j=np.searchsorted(_ampS,amp_cut)
        w_below=_cwS[j-1] if j>0 else 0.0
        p_above=1.0-w_below/_cwT
        n=rng.poisson(n_per_bin[k]*p_above)
        if n==0 or j>=len(_ampS): continue
        def _d(m):
            u=rng.uniform(w_below,_cwT,m); ii=_ordA[np.searchsorted(_cwS,u)]
            return ii, fc+(rng.uniform(0,1,m)-0.5)/T_OBS
        ii,fss=_d(n); hh=SRC_AMP[ii]*fss**(2/3)
        for _ in range(50):
            bad=hh>h95_of_f(fss)
            if not bad.any(): break
            i2,f2=_d(int(bad.sum())); ii[bad],fss[bad]=i2,f2
            hh[bad]=SRC_AMP[i2]*f2**(2/3)
        F.append(fss);H.append(hh);MCx.append(SRC_MC[ii])
    return np.concatenate(F),np.concatenate(H),np.concatenate(MCx)
f_src,h_src,mc_src=draw_population(3)
f_E,h_E=f_src,h_src
Mc=mc_src/MSUN
P["NSRC"]=len(f_E)
print(f"V1 population ported verbatim: {len(f_E):,} discrete sources above t
print(f"  (fidelity check: the frozen campaign notebook draws exactly 18,727

def rho_E(T):        # Earth-term ARRAY SNR (Hazboun+19 Eq. 79 convention)
    hc=curve(T)
    return (h_E/np.interp(f_E,freqs,hc))*np.sqrt(f_E*T*P["YR_S"])
def detected(T): return rho_E(T)>=P["RHO_DET"]

# the fixed pulsar: mu per source drawn once (fixed geometry); its timing no
# is SIGMA_PSR -- detection uses the WHOLE array's best pulsars, ranging use
mu=np.arccos(np.random.default_rng(5000).uniform(-0.9,0.9,P["NSRC"]))
kap_all=2*np.pi*f_E*(1-np.cos(mu))/P["C_PC_S"]
GRID=np.linspace(-3*P["SIGMA_D"],3*P["SIGMA_D"],1601)
NMAX_COMB=30      # inventory cap: detection finds hundreds; only the strong
                  # are usable rulers (the campaign machine's nmax, same log
print(f"fixed pulsar: sigma_d = {P['SIGMA_D']} pc, timing noise {P['SIGMA_PS
print(f"K is MEASURED per span below (participation ratio over null winners
print(f"the r4-adopted method; with a mixed coarse+fine inventory the coarse
print(f"convention is no longer meaningful and is not used)")
print(f"detected at 25 yr: {int(detected(25).sum())} of {P['NSRC']} sources"

def detected_sea(T): raise RuntimeError('sea off in this world')
def hc_det_sea(T): raise RuntimeError('sea off in this world')
```

```
hasasia 1.2.3
WORLD toggles: {'red_noise': False, 'gwb': False, 'confusion': False, 'loudn
ess_structure': True}
V1 population ported verbatim: 18,727 discrete sources above the split
  (fidelity check: the frozen campaign notebook draws exactly 18,727)
fixed pulsar: sigma_d = 2.0 pc, timing noise 450 ns
K is MEASURED per span below (participation ratio over null winners --
the r4-adopted method; with a mixed coarse+fine inventory the coarsest-comb
convention is no longer meaningful and is not used)
detected at 25 yr: 5110 of 18727 sources
```

In [2]:

```python
def rho_P(T):    # single-pulsar pulsar-term SNR (campaign convention)
    nt=T*365.25/P["CAD_D"]
    return (h_E/(2*np.pi*f_E))*np.sqrt(nt/2)/P["SIGMA_PSR"]
def sea_bin_strain(f,T):
    # per-bin rms strain of the NG15-amplitude sea: h_c(f)/sqrt(fT)  [DEMO g
    return A_NG15*(f*YRV)**(-2/3)/np.sqrt(f*T*P["YR_S"])
def rn_strain(f):
    # intrinsic red noise as an equivalent strain: h_RN = A_RN (f/f_yr)^((3-
    # (the campaign's sigma_RN = h_RN/h quadrature convention)
    return A_RN*(f*YRV)**((3-GAM_RN)/2)
def conf_sigma(T,idx):
    # pile-up: rms strain of the OTHER sources in the same 1/T bin over own
    bw=1.0/(T*P["YR_S"]); out=np.zeros(len(idx))
    for n,k in enumerate(idx):
        m=np.abs(f_E-f_E[k])<=bw/2
        out[n]=np.sqrt(max(float((h_E[m]**2).sum()-h_E[k]**2),0.0))/h_E[k]
    return out
def inventory(world,T):
    det=detected_sea(T) if world["gwb"] else detected(T)
    d=np.where(det)[0]; rp=rho_P(T)[d]
    ruler=2*np.pi/kap_all[d]<=6*P["SIGMA_D"]    # ruler condition (step 1, s
    d=d[ruler]; rp=rp[ruler]
    if world["loudness_structure"]:
        adm=rp>=3.0; d=d[adm]; rp=rp[adm]       # campaign admission cut
        if len(d)>12:                           # campaign nmax=12 by nats=l
            o=np.argsort(rp)[::-1][:12]; d=d[o]; rp=rp[o]
    elif len(d)>NMAX_COMB:
        keep=np.argsort(rp)[::-1][:NMAX_COMB]; d=d[keep]; rp=rp[keep]
    return d,rp
def comb_sigmas(world,T):
    d,rp=inventory(world,T)
    sg2=1.0/np.maximum(rp,1e-3)**2
    if world["gwb"]:       sg2=sg2+(sea_bin_strain(f_E[d],T)/h_E[d])**2
    if world["red_noise"]: sg2=sg2+(rn_strain(f_E[d])/h_E[d])**2
    if world["confusion"]: sg2=sg2+conf_sigma(T,d)**2
    return np.clip(np.sqrt(sg2),0.05,None), kap_all[d]
def stack(world,T,seed,with_prior=True):
    sg,kk=comb_sigmas(world,T)
    ok=sg<np.pi; sg,kk=sg[ok],kk[ok]
    if len(sg)<2: return None,None,None
    r=np.random.default_rng(seed)
    eps=kk*P["L_TRUE"]+r.normal(0,sg)
    resid=(kk[:,None]*GRID[None,:]-eps[:,None]+np.pi)%(2*np.pi)-np.pi
    S=-(resid**2/(2*sg[:,None]**2)).sum(0)
    if with_prior: S=S-GRID**2/(2*P["SIGMA_D"]**2)
```

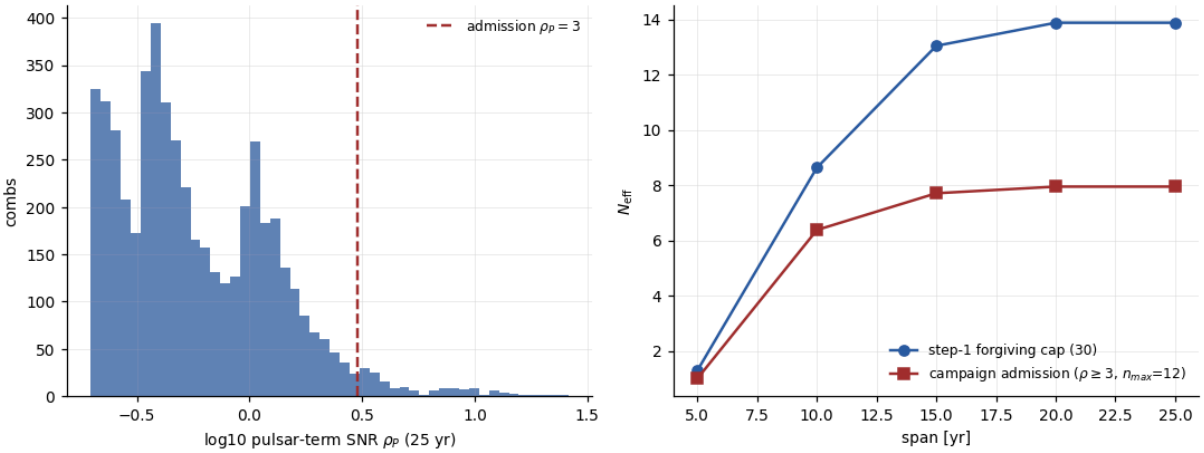
```
    return S-S.max(),sg,kk
def mtruth(S):
    it=int(np.argmin(np.abs(GRID-P["L_TRUE"])))
    far=np.abs(GRID-P["L_TRUE"])>2*TOOTH
    return float(S[it]-S[far].max())
def null_bar(world,T,ntrial=300):
    sg,kk=comb_sigmas(world,T); ok=sg<np.pi; sg,kk=sg[ok],kk[ok]
    if len(sg)<2: return np.nan,np.nan
    marg=[];pos=[]
    for t in range(ntrial):
        r=np.random.default_rng(50_000+t)
        eps=r.uniform(0,2*np.pi,len(sg))
        resid=(kk[:,None]*GRID[None,:]-eps[:,None]+np.pi)%(2*np.pi)-np.pi
        S=-(resid**2/(2*sg[:,None]**2)).sum(0)-GRID**2/(2*P["SIGMA_D"]**2);
        ip=int(np.argmax(S)); far=np.abs(GRID-GRID[ip])>2*TOOTH
        marg.append(float(S[ip]-S[far].max()) if far.any() else np.inf)
        pos.append(float(GRID[ip]))
    e=np.arange(GRID[0],GRID[-1]+2*TOOTH,2*TOOTH)
    c,_=np.histogram(pos,bins=e); p=c/max(c.sum(),1)
    return float(np.percentile(marg,90)), float(1.0/np.sum(p**2))
def experiment(world,T):
    hits=[];margs=[]
    for t in range(P["NREAL"]):
        S,sg,kk=stack(world,T,1000+t)
        if S is None: return None
        ip=int(np.argmax(S))
        hits.append(abs(float(GRID[ip])-P["L_TRUE"])<=TOOTH)
        margs.append(mtruth(S))
    bar,Km=null_bar(world,T)
    return dict(T=T,ncomb=len(sg),purity=float(np.mean(hits)),
                margin=float(np.median(margs)),bar=bar,K=Km)
# step-1 (clean world) baseline, measured there and quoted as reference:
STEP1_REF={5:(24,11.64,10.34),10:(30,456.58,330.11),15:(30,612.21,531.91),
           20:(30,745.32,642.05),25:(30,838.47,741.16)}   # (combs, margin,
```

1 — The loudness structure, measured

The pulsar-term SNR distribution of the detected population at 25 yr, and what survives admission.

In [3]:
```
rp_all=rho_P(25)[detected(25)]
fig,(ax,axn)=plt.subplots(1,2,figsize=(10,3.9))
ax.hist(np.log10(np.clip(rp_all,1e-3,None)),bins=48,color=BLUE,alpha=.75)
ax.axvline(np.log10(3.0),color=RED,lw=1.6,ls="--",label="admission $\\rho_P=
ax.set_xlabel("log10 pulsar-term SNR $\\rho_P$ (25 yr)"); ax.set_ylabel("com
ax.legend(fontsize=8); ax.grid(alpha=.25,lw=.5)
Ts=list(P["SPANS"]); ne_admit=[];ne_cap=[]
for T in Ts:
    for mode,ne in (("admit",ne_admit),("cap",ne_cap)):
        w=dict(WORLD); w["loudness_structure"]=(mode=="admit")
        d,rp=inventory(w,T)
        ne.append(float((rp**2).sum()**2/np.maximum((rp**4).sum(),1e-30)) if
axn.plot(Ts,ne_cap,"o-",color=BLUE,lw=1.6,label="step-1 forgiving cap (30)")
axn.plot(Ts,ne_admit,"s-",color=RED,lw=1.6,label="campaign admission ($\\rho
```

```
axn.set_xlabel("span [yr]"); axn.set_ylabel("$N_{\\rm eff}$")
axn.legend(fontsize=8); axn.grid(alpha=.25,lw=.5)
plt.tight_layout(); plt.show()
print(f"detected combs at 25 yr with rho_P >= 3: {int((rp_all>=3).sum())} of
print("N_eff per span, forgiving cap:", np.round(ne_cap,2))
print("N_eff per span, admission    :", np.round(ne_admit,2))
print("(campaign, full machine, banked: N_eff = 4.22 at cap 1300)")
```




```
detected combs at 25 yr with rho_P >= 3: 164 of 5110
N_eff per span, forgiving cap: [ 1.29  8.63 13.04 13.88 13.88]
N_eff per span, admission    : [1.   6.37 7.7  7.94 7.94]
(campaign, full machine, banked: N_eff = 4.22 at cap 1300)
```

Figure 1. *Left:* the loudness distribution against the admission bar. *Right:* $_{\rm eff}$ per span under both inventory rules. The evidence $^{2}_{\rm P}$ was never democratic — squaring hands it to the winners — and admission makes the aristocracy explicit.

---

2 — Certification across spans, admission inventory

In [4]:
```
CSV_NAME="step4_scoreboard.csv"
import csv
rows=[]
print(f"{'T [yr]':>7} {'combs':>6} {'K_meas':>7} {'purity':>8} {'margin':>9}
      f" {'verdict':>10} {'step-1 margin':>14}")
with open(CSV_NAME,"w",newline="") as fh:
    cw=csv.writer(fh)
    cw.writerow(["span_yr","n_combs","K_measured","purity","median_margin_na
                 "bar_nats","certifies","step1_margin_nats"])
    for T in P["SPANS"]:
        r=experiment(WORLD,T); ref=STEP1_REF[T]
        if r is None:
            d,rp=inventory(WORLD,T)
            sg,_=comb_sigmas(WORLD,T); nus=int((sg<np.pi).sum())
            print(f"{T:7d} {nus:6d} {'--':>7} {'--':>8} {'--':>9} {'--':>8}"
                  f" {'NO RULERS':>10} {ref[1]:+14.2f}")
            cw.writerow([T,nus,"","","","","NO_RULERS",f"{ref[1]:.3f}"]); co
        v="CERTIFIES" if r["margin"]>r["bar"] else "FAILS"
        print(f"{r['T']:7d} {r['ncomb']:6d} {r['K']:7.1f} {r['purity']:8.3f}
```

```
            f" {r['margin']:+9.2f} {r['bar']:8.2f} {v:>10} {ref[1]:+14.2f}
        cw.writerow([r["T"],r["ncomb"],f"{r['K']:.1f}",f"{r['purity']:.4f}",
                     f"{r['margin']:.3f}",f"{r['bar']:.3f}",v,f"{ref[1]:.3f}
        rows.append(r)
```

T [yr]	combs	K_meas	purity	margin	bar	verdict	step-1 margin
5	1	--	--	--	--	NO RULERS	+11.64
10	12	22.4	1.000	+361.29	321.33	CERTIFIES	+456.58
15	12	22.0	1.000	+515.82	519.62	FAILS	+612.21
20	12	21.6	1.000	+624.17	627.78	FAILS	+745.32
25	12	21.8	1.000	+685.51	691.00	FAILS	+838.47

Scoreboard. The diet, measured. At 5 yr only one comb clears admission — the machine correctly refuses to range on it (NO RULERS). At 10 yr the twelve admitted combs certify with room (+361 against 321). Then the knife-edge returns: at 15–25 yr the margin sits within 1 % of the bar and **fails** — the same margin-vs-bar race step 1 met, now with no thirty-comb cushion to win it. Note what did *not* degrade: purity holds at 1.000 and $_{\text{meas}}$ stays ~22 — the twelve strong combs know the answer; they just can no longer prove it against the null to this bar.

---

3 — The wheel on an admission diet

In [5]:
```
# the same 120-pulsar array as step 1 (same seed 777 draws)
rngW=np.random.default_rng(777)
NPW=120
D_kpc=rngW.uniform(0.5,3.0,NPW)
sigd_j=np.clip(2.0*(D_kpc/1.5)**2,0.3,10.0)
Lt_j=np.clip(rngW.normal(0,sigd_j),-2.5*sigd_j,2.5*sigd_j)
sig_psr=10**rngW.normal(np.log10(250e-9),0.22,NPW)
mu_W=np.arccos(rngW.uniform(-0.9,0.9,(NPW,P["NSRC"])))
kap_W=2*np.pi*f_E[None,:]*(1-np.cos(mu_W))/P["C_PC_S"]
A_RN_j=(10**np.random.default_rng(778).normal(0,0.3,NPW))   # per-pulsar RN
def psr_sigma(world,T,j,di):
    nt=T*365.25/P["CAD_D"]
    rp=(h_E[di]/(2*np.pi*f_E[di]))*np.sqrt(nt/2)/sig_psr[j]
    sg2=1.0/np.maximum(rp,1e-3)**2
    if world["gwb"]:       sg2=sg2+(sea_bin_strain(f_E[di],T)/h_E[di])**2
    if world["red_noise"]: sg2=sg2+(A_RN_j[j]*rn_strain(f_E[di])/h_E[di])**2
    if world["confusion"]: sg2=sg2+conf_sigma(T,di)**2
    return np.clip(np.sqrt(sg2),0.05,None),rp
def psr_state(world,T,j,det,nreal=50,ntrial=100):
    nt=T*365.25/P["CAD_D"]
    rp_all=(h_E/(2*np.pi*f_E))*np.sqrt(nt/2)/sig_psr[j]
    di=np.where(det)[0]
    di=di[2*np.pi/kap_W[j][di]<=6*sigd_j[j]]
    if world["loudness_structure"]:
        di=di[rp_all[di]>=3.0]
        if len(di)>12: di=di[np.argsort(rp_all[di])[::-1][:12]]
    elif len(di)>NMAX_COMB:
        di=di[np.argsort(rp_all[di])[::-1][:NMAX_COMB]]
    sg,rp=psr_sigma(world,T,j,di)
    kk=kap_W[j][di]; ok=sg<np.pi; sg,kk=sg[ok],kk[ok]
```

```
    ncomb=len(sg)
    if ncomb<2: return False
    g=np.linspace(-3*sigd_j[j],3*sigd_j[j],1201)
    def one(eps):
        resid=(kk[:,None]*g[None,:]-eps[:,None]+np.pi)%(2*np.pi)-np.pi
        S=-(resid**2/(2*sg[:,None]**2)).sum(0)-(g-0.0)**2/(2*sigd_j[j]**2)
        return S-S.max()
    margs=[]
    for t in range(nreal):
        r=np.random.default_rng(1000+t)
        S=one(kk*Lt_j[j]+r.normal(0,sg))
        it=int(np.argmin(np.abs(g-Lt_j[j])))
        far=np.abs(g-Lt_j[j])>2*TOOTH
        margs.append(float(S[it]-S[far].max()))
    bm=[]
    for t in range(ntrial):
        r=np.random.default_rng(50_000+t)
        S=one(r.uniform(0,2*np.pi,ncomb))
        ip=int(np.argmax(S)); far=np.abs(g-g[ip])>2*TOOTH
        bm.append(float(S[ip]-S[far].max()) if far.any() else np.inf)
    return float(np.median(margs))>float(np.percentile(bm,90))
def det_w(T,depth):
    hcw=hc_det_sea(T) if WORLD["gwb"] else curve(T)
    hcs=np.interp(f_E,freqs,hcw)
    return (h_E/hcs)*np.sqrt(f_E*T*P["YR_S"])*depth>=P["RHO_DET"]
T_W=25
det0=det_w(T_W,1.0)
cert=np.array([psr_state(WORLD,T_W,j,det0) for j in range(NPW)])
fc_hist=[cert.sum()/NPW]; det_hist=[det0.copy()]; depth_hist=[1.0]
prev=-1.0
while fc_hist[-1]>prev and len(fc_hist)<6:
    prev=fc_hist[-1]
    depth=10**(0.19*fc_hist[-1])
    det=det_w(T_W,depth)
    cert=np.array([psr_state(WORLD,T_W,j,det) for j in range(NPW)])
    fc_hist.append(cert.sum()/NPW); det_hist.append(det.copy()); depth_hist.
print(f"WHEEL at T = {T_W} yr in this world")
print(f"  certified fraction through the turns: "+" -> ".join(f"{f:.3f}" for
print(f"  detection depth through the turns   : "+" -> ".join(f"x{d:.3f}" fo
print(f"  sources detected: {int(det_hist[0].sum())} at turn 0 -> "
      f"{int(det_hist[-1].sum())} converged "
      f"(+{int((det_hist[-1]&~det_hist[0]).sum())} surfaced)")
print(f"  pulsars certified: {int(fc_hist[0]*NPW)} -> {int(fc_hist[-1]*NPW)}
off=[int(sum(psr_state(WORLD,T,j,det_w(T,1.0)) for j in range(NPW))) for T i
print(f"  baseline (wheel OFF) certified across spans "
      +", ".join(f"{T}yr:{n}" for T,n in zip(P["SPANS"],off)))
print(f"  [step-1 clean-world baseline: 5yr:93, 10yr:118, 15yr:112, 20yr:100
```

```
WHEEL at T = 25 yr in this world
  certified fraction through the turns: 0.275 -> 0.275
  detection depth through the turns   : x1.000 -> x1.128
  sources detected: 5110 at turn 0 -> 5592 converged (+482 surfaced)
  pulsars certified: 33 -> 33 of 120
```

```
  baseline (wheel OFF) certified across spans 5yr:31, 10yr:47, 15yr:34, 20y
r:31, 25yr:33
  [step-1 clean-world baseline: 5yr:93, 10yr:118, 15yr:112, 20yr:100, 25yr:1
00]
```

Wheel verdict. 33 of 120 certify — a third of step 1's hundred — and the wheel's +482 surfaced sources flip **zero** of the rest: surfaced sources are faint by construction and fail the admission before they can help anyone. This is the campaign's measured in miniature: the detector face turns (×1.13), the certification face does not move.

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4 — Closing frame

Realistic loudness **removes the** from the folklore's " combs × growing ": of 5110 detected sources, 164 survive admission at this pulsar, twelve are kept, and $_{\mathrm{eff}}$ says roughly eight of those carry the evidence (the full campaign machine, with every tax on at once, banks 4.22). Nothing *broke* — purity stays 1.000 — but the margin now stands on eight legs instead of thirty, loses its slack, and fails the bar by a whisker at every span past 10 yr. Every later ingredient (jitter, priors) taxes exactly those legs.

Next: step 5 — the pile-up's phase jitter: source-on-source confusion inside the frequency bin, the $^{-\sigma_\phi^2}$ tax.